

Vellacombe Discovery Ltd

Phenoxyethyl amides: protected intermediate and polar-tail delivery

Route scout and delivery — final report, revision A

Programme: VD-17-044

Accession: VA20170410

Client: Rennelwick Institute Drug Discovery Unit

Date: 02 Aug 2017

Author: Dr R. Calderwood, Head of Medicinal Chemistry

1 Final material schedule

Rennelwick Institute Drug Discovery Unit received 24 compounds under VD-17-044. Analogue preparation began on 24 Apr 2017; the interim checkpoint was 12 Jun 2017, and chemistry completion was recorded on 01 Aug 2017. This final Route scout and delivery report is issued by Dr R. Calderwood on 02 Aug 2017.

The protected aminoethyl intermediate opened access to the requested polar-tail variants without disrupting amide coupling. Hassan Darwish used that common intermediate to separate the coupling problem from access to the tail functionality. Ruth Calderwood retained final identity and isolation as the route gate. The advantage is chemoselective access in the reported route, not a claim that every protecting group or unmade tail behaves equivalently.

Table 1 brings the identifier, assigned structure, molecular weight, isolated yield, UV area purity and amount into the same entry. Only this schedule constitutes the reported delivery. Suggested structures elsewhere in the report are not additional isolated compounds.

Table 1. Compounds delivered; final isolated yield, HPLC purity in UV area percent at 254 nm and amount in mg.

Compound	SMILES	MW	Yield (%)	HPLC purity (UV area %, 254 nm)	Amount (mg)
RI13-02900	<chem>O=C(NCOC1ccc(-c2ccncc2)cc1)c1cnc(NC2CCCCC2)c1</chem>	402.5	7	95.7	38
RI13-02910	<chem>O=C(NCOC1ccc(-c2cncnc2)cc1)c1cnc(N2CCOCC2)c1</chem>	391.4	4	96.2	29
RI13-02912	<chem>CN1CCN(c2cc(C(=O)NCOc3ccc(-c4cncnc4)cc3)ccn2)CC1</chem>	404.5	5	98.6	9
RI13-02918	<chem>CCN1CCN(c2cc(C(=O)NCOc3ccc(-c4cccnc4)cc3)ccn2)CC1</chem>	417.5	31	99.4	33
RI13-02930	<chem>Cc1ccc(-c2ccc(OCNC(=O)c3ccnc(OC4CCN(C)CC4)c3)cc2)nc1</chem>	432.5	21	97.2	19
RI13-02934	<chem>O=C(NCOC1ccc(-c2cncnc2)cc1)c1cnc(-c2ccncc2)c1</chem>	383.4	37	97.3	41
RI13-02936	<chem>CN1CCN(c2cc(C(=O)NCOc3ccc(-c4cccnc4)cc3)ccn2)CC1</chem>	403.5	32	95.7	35
RI13-02946	<chem>CN1CCN(c2cc(C(=O)NCOc3ccc(-c4ccncc4)cc3)ccn2)CC1</chem>	403.5	41	97.0	9
RI13-02988	<chem>CCN1CCN(c2cc(C(=O)NCOc3ccc(-c4ccc(C)cn4)cc3)ccn2)CC1</chem>	431.5	24	95.4	33
RI13-02996	<chem>Cc1ccc(-c2ccc(OCNC(=O)c3ccnc(N4CCOC4)c3)cc2)nc1</chem>	404.5	40	97.2	11
RI13-03002	<chem>CCN1CCN(c2cc(C(=O)NCOc3ccc(-c4ccc(NC(C)=O)cc4)cc3)ccn2)CC1</chem>	473.6	40	95.4	19
RI13-03006	<chem>N#Cc1ccc(-c2ccc(OCNC(=O)c3ccnc(N4CCCC4)c3)cc2)cc1</chem>	398.5	19	96.6	32
RI13-03008	<chem>N#Cc1ccc(-c2ccc(OCNC(=O)c3ccnc(-c4ccncc4)c3)cc2)cc1</chem>	406.4	54	94.9	25
RI13-03014	<chem>O=C(NCOC1ccc(-c2ccncc2)cc1)c1cnc(NC2CCCCC2)c1</chem>	402.5	44	97.5	49
RI13-03018	<chem>Cc1ccc(-</chem>	388.5	23	95.3	27

	<chem>c2ccc(OCNC(=O)c3ccnc(N4CCCC4)c3)cc2)nc1</chem>				
RI13-03020	<chem>CC(=O)Nc1ccc(-c2cc(C(=O)NCOc3ccc(-c4ccc(C)cn4)cc3)ccn2)cc1</chem>	452.5	53	93.2	40
RI13-03038	<chem>Nc1ccc(-c2ccc(OCNC(=O)c3ccnc(N4CCOC4)c3)cc2)cc1</chem>	404.5	15	98.2	9
RI13-03048	<chem>O=C(NCOc1ccc(-c2ccncc2)cc1)c1ccnc(-c2ccncc2)c1</chem>	382.4	46	95.2	16
RI13-03050	<chem>CC(=O)Nc1ccc(-c2cc(C(=O)NCOc3ccc(-c4ccncc4)cc3)ccn2)cc1</chem>	438.5	84	95.9	31
RI13-03054	<chem>COc1ccc(-c2ccc(OCNC(=O)c3ccnc(N4CCN(C)CC4)c3)cc2)cc1</chem>	432.5	51	97.1	31
RI13-03056	<chem>Cc1ccc(-c2ccc(OCNC(=O)c3ccnc(-c4ccncc4)c3)cc2)nc1</chem>	396.5	56	96.3	57
RI13-03078	<chem>O=C(NCOc1ccc(-c2ccncc2)cc1)c1ccnc(N2CCOCC2)c1</chem>	390.4	22	96.6	26
RI13-03086	<chem>CC(=O)Nc1ccc(-c2ccc(OCNC(=O)c3ccnc(-c4ccncc4)c3)cc2)cc1</chem>	438.5	42	99.2	29
RI13-03090	<chem>CC(=O)Nc1ccc(-c2ccc(OCNC(=O)c3ccnc(-c4ccc(NC(C)=O)c4)c3)cc2)cc1</chem>	494.6	59	98.7	52

2 Remit, chronology and responsibility

The remit was Analogue preparation for the RI13 set. The report covers route decisions affecting access to material and the identity basis needed for the client comparison. The programme turnaround was 99 days. That figure describes the programme interval, not a reaction time or assay incubation.

Dr R. Calderwood led the programme, Dr H. Darwish directed synthetic execution and Clara Fenwick led analytical release. The final gate closed on 01 Aug 2017; this report follows completion rather than reopening the route as a draft. Programme accession VA20170410 links the material schedule to the benchmark below.

Earlier intermediates remain useful experimental context but are not counted again as delivered compounds. A repeat request should name the released identifier and the chemistry question. The final schedule is the endpoint for reconciling the set; individual compound cards remain the source for recovering a preparation.

Table 2. Programme benchmark and administrative reconciliation.

Metric	Value
Project	VD-17-044
Client	Rennelwick Institute Drug Discovery Unit
Compounds tested	24
Series / chemotype	Rennelwick phenoxyethyl amide series 13 (Phenoxyethyl amide)
Assays	Custom synthesis and analogue preparation
Turnaround (calendar days)	99
Price	GBP 34,300

3 Chemistry hypothesis

The polar-tail programme required access to phenoxyethyl amides while retaining control of which function participated in amide formation. The hypothesis was that a protected aminoethyl intermediate would allow the desired coupling without interference from another reactive function. Protection was a route tool, not a final pharmacological feature. The intended outcome was an isolated set with the requested tails; protected precursors were not completed delivery compounds.

The programme was organised around material needed to answer the client question. Route success was judged at the isolated product, not at an attractive intermediate or plausible scheme. This distinction limits the final claims: chemistry establishes access and identity for the delivered set but supplies no biological ranking that has not been measured here. The proposed follow-up must retain the same distinction.

4 Route gate and sequence decision

The protected aminoethyl intermediate opened access to the requested polar-tail variants without disrupting amide coupling. Hassan Darwish used that common intermediate to separate the coupling problem from access to the tail functionality. Ruth Calderwood retained final identity and isolation as the route gate. The advantage is chemoselective access in the reported route, not a claim that every protecting group or unmade tail behaves equivalently.

The interim discussion on 12 Jun 2017 was a route checkpoint, not the delivery endpoint. By 01 Aug 2017, the completed material schedule could be reconciled with the programme. A repeat chemist should preserve the reason for the route choice as well as the sequence. Repeating an operation without knowing which recovery or identity risk it controlled can recreate the original difficulty.

5 Analogue preparation

A protected intermediate simplifies a convergent operation but introduces a later removal step. The intended connection must remain identifiable through coupling, and the protected precursor must stay distinct from the final product. The compound card links those stages so an analytical entry cannot attach to the wrong drawing. Tail variants are isolated products only where they appear in Table 1; other combinations remain proposals despite plausible access through the same intermediate.

For repeat work, follow the compound card for the chosen identifier and reconcile the intended product with Table 1. A common scaffold is not a common experimental outcome. Substitution changes can affect work-up, purification and the evidence needed for assignment. No unprepared variant is claimed to have crossed the route gate merely because its drawing resembles a delivered compound.

6 Separation, work-up and recovery

Polar-tail products can behave differently from their protected intermediates during extraction and purification. Attribute recovery to the material actually isolated rather than assuming common behaviour across the set. Table 1 gives final yields and amounts, not a uniform deprotection recovery or coupling conversion. A cleaner final profile also does not establish removal of every precursor unless identity and the relevant impurity checks are considered together.

Isolated yield is a route measure and delivered amount is a material measure. Both matter because the comparison requires the actual compound in hand. Neither column tells a chemist where loss occurred. Return to intermediate and fraction entries for that question. An estimate derived only from the final table would conceal the difference between reaction performance and recovery.

7 Identity and assignment

Clara Fenwick handled release with the delivered compound distinguished from the protected route intermediate. The final structure field is not permission to test a precursor under the same identifier. Where available amine functions change during the route, the identity basis must support the intended amide connection. Nominal formula alone cannot demonstrate which function coupled. The route history and final analytical entry must remain linked to the delivery identifier.

The release entry must stay connected to the isolate. A structure redrawn for a client presentation should retain the chemistry identifier so the assignment basis can be recovered. This report does not reproduce raw spectra or supply invented peak assignments. It records the programme conclusion and explains where the underlying experimental history is retained.

8 Purity statement and analytical limits

The HPLC values in Table 1 are UV area percent at 254 nm. They describe the reported chromatographic profile under that detection basis. They are not weight percent, solvent content or a direct assay of active parent in a weighed portion. The amount field is expressed in mg and must remain separate from purity when planning the comparison.

Clara Fenwick led analytical release. Identity and chromatographic purity answer complementary questions: a dominant peak cannot independently establish the assigned connection, position or configuration. An assigned structure also does not remove the need to consider impurities. The schedule retains the available numerical values without adding a universal purity threshold or an unstated correction to the delivered amounts.

A client query concerning precipitation, unexpected peaks or identity should return against the delivery identifier. The compact table is a reconciliation aid, not a substitute for the analytical entry needed to investigate a particular sample.

9 Isolated yield and material accounting

Final yields and amounts remain separated by compound. No pooled recovery or campaign mean has been calculated. Such a figure could blur route differences and imply a common experimental basis that the delivery table does not establish. The practical question is whether identifier, structure and amount refer to the same released material.

Compound cards may contain intermediate masses, recovered fractions and material held for further work. Those entries do not automatically belong to the client delivery. Use Table 1 for the delivered set and the route scout for an individual recovery question. A request for more material must distinguish repeat product from a new proposed analogue.

The programme closed with 24 delivered compounds. This count is not enlarged by an intermediate, an alternative drawing or a proposed comparison. Detailed weighing and transfer history belong to the individual chemistry entries; the final schedule supplies report-level reconciliation.

10 Client hand-off and identifier continuity

The RI13 identifiers connect isolated chemistry to the client comparison. Use VD-17-044 and accession VA20170410 for clarification. Assigned structures help reconcile a vial label with its intended product, but the delivery identifier remains the primary link to prepared material.

If a client transfers material for biological preparation, retain the original identifier and distinguish that later preparation description from chemistry release. Replacing an identifier with a proposed structure name risks merging different samples under a general scaffold label. The risk matters particularly where apparently similar structures address different chemical questions.

Chemistry completion was recorded on 01 Aug 2017. This report supplies no unstated courier date, receipt signature or storage condition. Dr H. Darwish is the route contact; Clara Fenwick is the analytical reconciliation contact; Dr R. Calderwood retains responsibility for the overall chemistry conclusion.

11 Interpretation for the client comparison

Rennelwick Institute Drug Discovery Unit received the requested polar-tail variants through the protected aminoethyl route. No nuclear receptor response or binding values are supplied here. The chemistry outcome is access without sacrificing control of amide formation. Later interpretation should retain the actual delivery identifier and distinguish a tail change from a change in sample handling. A proposed combination must not inherit an apparent biological result from its precursor.

The useful follow-up is a comparison whose members can be traced to released material. Keep the intended chemical difference explicit. Do not combine a proposed product and an isolated compound within a result entry. A later biological trend may justify new chemistry, but cannot change retrospectively which material was delivered.

This is a chemistry close-out. It provides no invented response values, selectivity ratios or potency rankings. Programme turnaround is not an assay setting, and isolated yield is not an indicator of target activity.

12 Next analogue and carried route risk

The next proposed analogue changes a single polar-tail feature while retaining the established amide connection and protected-intermediate logic. It is outside this delivery. The route risk is altered compatibility with protection removal or loss transferred into the final work-up. Address that risk in a small route scout before assuming the common intermediate guarantees delivery. Extension requires its own final identity and isolated material.

A repeat route scout should identify the operation controlling the original risk and the evidence that closed it. Do not omit that gate because the final compound looks familiar. Substituent changes and material history can affect recovery without changing the intended chemical question. Preserve the link between the proposed route and the actual isolated outcome.

The report does not qualify a working scale beyond the delivered chemistry or impose new process conditions. Recover charge quantities, timings and purification parameters from the relevant experimental entry. A change in the established sequence needs its own reason and compound-card history.

13 Follow-up decision and status

The next analogue described above is proposed chemistry. It has no additional delivery entry, assigned yield, purity, amount or biological result in this report. Its purpose is a defined extension of the successful route while retaining the chemical distinction that motivated the programme.

Dr R. Calderwood recommends that Rennelwick Institute Drug Discovery Unit reconcile the intended comparison with the released identifiers before expanding preparation. Once the chemical question is selected, the next route scout can be scoped around it. A broad set assembled before that choice may add material without clarifying interpretation.

The gate for a new analogue remains identifiable isolated material. A convincing scheme is not a completed delivery. This final report closes the present programme and proposes a reasoned next step without presenting it as work already performed.

Appendix A — Delivery-table conventions

Compound is the delivery identifier linking material to chemistry. SMILES is the compact assigned-structure field. MW is the reported molecular-weight field associated with it. These fields must not be used to infer an unstated counter-ion, solvate or assignment detail. The drawing summarises identity; the compound card preserves its basis.

Yield (%) is the reported isolated yield. HPLC purity is UV area percent at 254 nm. Amount (mg) is the reported isolated amount. Keep the bases separate. The table supplies no raw peak areas, moisture correction, uncertainty estimate or across-programme normalisation. It also does not substitute for a preparation-specific experimental entry.

To reproduce a delivered product, select the identifier, recover its compound card and reconcile the final isolate with its route history. To extend the chemistry, create a route proposal rather than repurposing an existing delivery entry for an unmade structure.

Appendix B — Experimental history and repeat instructions

Compound cards connect the protected aminoethyl intermediate, amide coupling and final tail-bearing product. Consult the relevant card before repeating a preparation; Table 1 does not supply protection-removal conditions or intermediate quantities. This report avoids inventing them. Reconcile the intermediate identity, intended coupling function and final isolation before issuing repeat material. Record a changed protection strategy as a new route choice even if the proposed final structure remains unchanged.

A repeat request should identify the compound, why additional material is needed and the route question to preserve. The chemist can then recover the operation sequence and compare the repeat with the established identity basis. A broad scaffold description is insufficient when the conclusion depends on separation, selective access or assignment.

The report remains the final account of delivered chemistry. It does not overwrite experimental entries or create replacement conditions. If subsequent preparation departs from the established route, retain that departure in its own compound card and distinguish the new material history from the present delivery.

14 Final conclusion and authorisation

The protected aminoethyl intermediate opened access to the requested polar-tail variants without disrupting amide coupling. Hassan Darwish used that common intermediate to separate the coupling problem from access to the tail functionality. Ruth Calderwood retained final identity and isolation as the route gate. The advantage is chemoselective access in the reported route, not a claim that every protecting group or unmade tail behaves equivalently.

The final delivery comprises 24 isolated compounds for Rennelwick Institute Drug Discovery Unit. Amounts, isolated yields and UV area purity are recorded in Table 1. The route supports the reported client comparison; proposed follow-up structures remain outside the delivery. No biological ranking is assigned in this conclusion.

Issued on 02 Aug 2017, after completion on 01 Aug 2017. Dr R. Calderwood authorises the chemistry conclusion. Dr H. Darwish is identified for synthetic execution and Clara Fenwick for analytical release. Revision A is final and has no open comments or provisional replacements for the material schedule.

Name	Role	Signature	Date
Dr R. Calderwood	Head of Medicinal Chemistry — final report authorisation	<u>D Calderwood</u>	02 Aug 2017